



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 – 2026-2027 | Trimester - 4

Project Abstract Submission Form

Sec No.	Team No.	Project Title
7	17	Lost & Found Item Tracker for Campus

1. Team Details

Roll Number(s)	Name of the Student	Signature
2520030237	Hansika Reddy Gunapati	
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2. Frameworks / Platforms / Software Used

- **Android (Kotlin)** – native mobile application for students, security desk and admin
- **Firebase Firestore** – primary NoSQL distributed database for items, users and claims
- **Firebase Authentication** – campus email based user login and role management
- **Firebase Cloud Storage** – storage of lost/found item images
- **Firebase ML Kit (Image Labeling)** – automatic tagging of item photos for matching
- **Firebase Cloud Messaging (FCM)** – push notifications for potential matches and claim updates
- **Firebase Cloud Functions** – server-side matching engine and validation logic
- **Android Studio, Git & GitHub** – development environment and version control

3. CO – Module Mapping Summary

CO	Project Module	Core Feature(s) Delivered
CO1	System Architecture & Request Lifecycle	Report Lost Item, Post Found Item (end-to-end request flow)
CO2	Database Design & ER Modelling	Normalized schema for Users, Items, Claims
CO3	Query & Matching Engine	AI Matching Engine, Admin Moderation dashboard
CO4	Claim Verification & Transactions	Chat & Claim Verification (atomic claim confirmation)
CO5	Search & Indexing	Fast category/location search, ranked matches
CO6	Scalability & Distributed Backend	Notifications (FCM fan-out), multi-campus scaling

4. Abstract

Lost-and-found handling on most campuses is still manual: items are logged on notice boards, in group chats, or at the security desk, with no searchable record and no automated way to connect a lost report with a matching found post. Recovery rates stay low and the entire moderation burden sits with campus staff.

This project, **Lost & Found Item Tracker for Campus**, addresses this gap with an Android (Kotlin) application backed by Firebase. Students can report a lost item or post a found item along with a photo, category and location. An AI chatbot cross-checks the two descriptions and images to verify whether a reported lost item and a found item are genuinely the same, using ML Kit image labels combined with keyword and category similarity to rank likely matches. Users can search and filter items by category, location, date, and attributes such as size, shape and colour, and can look up matches even against items that were posted long before the search — not just recent ones.

Once a likely match is found, an in-app chat feature lets the two parties communicate directly to verify ownership before a claim is confirmed. Claim confirmation is wrapped in a transaction boundary so that two users can never simultaneously claim the same item, directly addressing concurrency anomalies. Real-time notifications and alerts inform users the moment a potential match, claim request, or status update occurs. An admin oversees the platform, moderating disputed or suspicious posts and stepping in whenever a claim is contested.

The backend is organised as layered request handling, validation, service and persistence layers, with a normalized data model for users, item reports and claims that avoids update anomalies. Category, location and date fields are indexed to keep search fast as the item log grows, and Firestore's distributed, replica-backed architecture supports read scaling and availability as the system grows across the campus.

Core modules include Report Lost Item, Post Found Item, an AI Matching & Verification Chatbot, Chat & Claim Verification, Real-Time Notifications & Alerts, Admin Dispute Moderation, and Advanced Search & Filtering.

The system is also designed with growth in mind: as more students and campuses come on board, the underlying Firestore backend is expected to scale horizontally, with campus-wise partitioning planned so that each off-campus location's item reports remain fast to query even as overall volume increases. The notification and chat paths favour availability, ensuring the app stays responsive even under heavy concurrent use, while the claim-confirmation path stays strictly consistent so that ownership disputes never arise from a race condition. Taken together, the project demonstrates how sound database design, layered backend architecture, and lightweight AI verification can combine to turn a manual, error-prone campus process into a reliable, self-service system. The result is a faster, auditable, and largely self-service lost-and-found workflow for the campus community — one that reduces the burden on security staff while giving students a transparent, trustworthy way to recover their belongings.

Guide Name and Signature (remarks if any):